

IIM — Inference Integrity Module

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Independent Methodological Authority

OriginID: OOF-OID-AI-IIM-2026-06-03-0001

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: Cognitive Governance Intelligence Architecture (CLIA®)

Operational Layer: Cognitive Reasoning Governance Layer

Governed Space: Inference Integrity

Category: AI & Interpretation

Subcategory: Inference Governance

Type: Cognitive Reasoning Integrity Module

Parent Standard: Cognitive Reasoning Integrity Standard (CRIS)

Version: 1.0

Status: Canonical · Open Module

Origin Date: 3 June 2026

Compatibility: OOF Methodology OS ·

Cognitive Reasoning Integrity Standard (CRIS) ·

Cognitive Interpretation Integrity Standard (CIIS) ·

Cognitive Integrity Standard (CIS) ·

Multi-Layer Truth Validation Framework (MTVF) ·

INTEGROS® — Integrity Standard

AI-Readable: Yes

Authority: OOF

Protection: MIP — Methodological Intellectual Property

Canonical Language: English (UCL)

Canonical Definition

Inference Integrity Module (IIM) defines the structural conditions under which inference transitions, reasoning steps, logical progressions, intermediate conclusions, and cognition-path dependencies remain traceable, coherent, reconstructable, governance-valid, and operationally reliable throughout reasoning processes.

IIM governs inference.

The module ensures that cognition remains capable of moving from one reasoning state to another without introducing hidden logical discontinuities, unsupported transitions, inference distortion, or governance-invalid reasoning pathways.

A system satisfies IIM only if:

- inference transitions remain traceable
- reasoning steps remain reconstructable
- intermediate conclusions remain visible
- inference dependencies remain identifiable
- unsupported transitions remain detectable

- reasoning progression remains operationally valid

A cognition environment that cannot explain how conclusions emerged from preceding reasoning states does not satisfy IIM.

Module Operational Role

IIM defines the inference-governance layer of CRIS.

Its role is to preserve valid logical movement between reasoning states.

Module Operational Space

IIM governs:

- inference transitions
- reasoning progression
- logical movement
- intermediate conclusions
- reasoning pathways
- inference traceability
- cognition-step integrity
- reasoning continuity

The module applies wherever cognition generates conclusions through multi-step reasoning.

Module Function

The module applies wherever systems must preserve:

- traceable reasoning
- valid inference transitions
- reconstructable cognition pathways
- governance-valid reasoning progression
- logical continuity
- accountable conclusion formation

Its function is to ensure that reasoning remains visible between starting assumptions and final conclusions.

Minimum Implementation Framework

1. Define the Inference Object

The organization must define which inference processes require governance.

This may include:

- analytical reasoning
 - planning reasoning
 - strategic reasoning
 - predictive reasoning
 - operational reasoning
 - diagnostic reasoning
 - investigative reasoning
 - autonomous agent reasoning
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2. Define Inference Integrity Conditions

The system must define the conditions under which inference remains valid.

This includes:

- transition-validity requirements
- reasoning-traceability requirements
- continuity requirements
- reconstruction requirements
- justification requirements
- governance-valid inference conditions

3. Define Inference Degradation Detection Logic

The system must define how inference degradation is identified.

This may include:

- unsupported transitions
- missing reasoning steps
- inference discontinuities
- hidden assumptions
- logical gaps
- reasoning inconsistencies

4. Define Operational Response or Governance Logic

The system must define governance logic for inference-integrity failures.

Governance response may include:

- reasoning review
- inference reconstruction
- logical validation
- governance intervention
- escalation
- conclusion restriction
- operational invalidation where required

5. Preserve Traceability & Restrict Invalid Conditions

The system must preserve reconstructable traceability of:

- reasoning steps
- inference transitions
- intermediate conclusions
- governance reviews
- validation actions
- resulting conclusions

A cognition environment must not remain inference-valid if materially significant reasoning transitions cannot be reconstructed, explained, or governed.

Use Case 1 — Autonomous Strategic Planning Agent

Scenario

An autonomous planning agent evaluates operational information and generates strategic recommendations for future actions.

Application

IIM governs reasoning progression, inference transitions, and intermediate conclusions throughout planning activities.

Result

The organization gains stronger reasoning transparency, improved conclusion reliability, and reduced risk of hidden inference failures.

Use Case 2 — Intelligence Analysis Environment

Scenario

An intelligence system analyzes multiple information sources and generates assessments regarding future risks and opportunities.

Application

IIM governs how conclusions emerge from evidence through traceable inference pathways.

Result

The environment gains stronger analytical accountability, improved reasoning visibility, and reduced exposure to unsupported conclusions.

Canonical Closing Statement

Inference Integrity Module (IIM) defines the structural conditions under which inference transitions, reasoning steps, logical progressions, intermediate conclusions, and cognition-path dependencies remain traceable, coherent, reconstructable, governance-valid, and operationally reliable throughout reasoning processes.

Conclusions cannot remain valid if inference becomes invisible.

Inference integrity therefore becomes a foundational integrity condition of governable reasoning.

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Canonical Source

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